

A Practical Guide to Semantic segmentation

TOPIC -- SEMANTIC SEGMENTATION

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Groups of image segmentation Semantic segmentation is an approach detecting, for every pixel, the belonging class. For example, in a figure with many people, all the pixels belonging to persons will have the same class id and the pixels in the background will be classified as background. Instance segmentation is an approach that identifies, for every pixel, the specific belonging instance of the object. It detects each distinct object of interest in the image. For example, when each person in a figure is segmented as an individual object. Panoptic segmentation combines both semantic and instance segmentation. Like semantic segmentation, panoptic segmentation is an approach that identifies, for every pixel, the belonging class. Moreover, like in instance segmentation, panoptic segmentation distinguishes different instances of the same class.

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Isolated Point Detection The detection of isolated points in an image is a fundamental part of image segmentation. This process primarily depends on the second derivative, indicating the use of the Laplacian operator. The Laplacian of a function $f(x, y)$ is given by:

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Variational methods The goal of variational methods is to find a segmentation which is optimal with respect to a specific energy functional. The functionals consist of a data fitting term and a regularizing terms. A classical representative is the Potts model defined for an image f by

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$$i = \begin{cases} i_{\text{new}}, & \text{if } \Delta U \leq 0 \\ i_{\text{new}}, & \text{if } \Delta U > 0 \text{ and } \delta < e^{-\Delta U / T} \\ i_{\text{old}}, & \text{otherwise} \end{cases}$$

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$$\frac{\partial^2 f(x, y)}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

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Optimization algorithms Each optimization algorithm is an adaptation of models from a variety of fields and they are set apart by their unique cost functions. The common trait of cost functions is to penalize change in pixel value as well as difference in pixel label when compared to labels of neighboring pixels.

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Fast marching methods The fast marching method has been used in image segmentation, and this model has been improved (permitting both positive and negative propagation speeds) in an approach called the generalized fast marching method.

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Partial differential equation-based methods Using a partial differential equation (PDE)-based method and solving the PDE equation by a numerical scheme, one can segment the image. Curve propagation is a popular technique in this category, with numerous applications to object extraction, object tracking, stereo reconstruction, etc. The central idea is to evolve an initial curve towards the lowest potential of a cost function, where its definition reflects the task to be addressed. As for most inverse problems, the minimization of the cost functional is non-trivial and imposes certain smoothness constraints on the solution, which in the present case can be expressed as geometrical constraints on the evolving curve.

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Parametric methods Lagrangian techniques are based on parameterizing the contour according to some sampling strategy and then evolving each element according to image and internal terms. Such techniques are fast and efficient, however the original "purely parametric" formulation (due to Kass, Witkin and Terzopoulos in 1987 and known as "snakes"), is generally criticized for its limitations regarding the choice of sampling strategy, the internal geometric properties of the curve, topology changes (curve splitting and merging), addressing problems in higher dimensions, etc.. Nowadays, efficient "discretized" formulations have been developed to address these limitations while maintaining high efficiency. In both cases, energy minimization is generally conducted using a steepest-gradient descent, whereby derivatives are computed using, e.g., finite differences.

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Dual clustering method This method is a combination of three characteristics of the image: partition of the image based on histogram analysis is checked by high compactness of the clusters (objects), and high gradients of their borders. For that purpose two spaces have to be introduced: one space is the one-dimensional histogram of brightness $H = H(B)$; the second space is the dual 3-dimensional space of the original image itself $B = B(x, y)$. The first space allows to measure how compactly the brightness of the image is distributed by calculating a minimal clustering k_{min} . Threshold brightness T corresponding to k_{min} defines the binary (black-and-white) image – bitmap $b = \phi(x, y)$, where $\phi(x, y) = 0$, if $B(x, y) < T$, and $\phi(x, y) = 1$, if $B(x, y) \geq T$. The bitmap b is an object in dual space. On that bitmap a measure has to be defined reflecting how compact distributed black (or white) pixels are. So, the goal is to find objects with good borders. For all T the measure $MDC = G/(k \times L)$ has to be calculated (where k is difference in brightness between the object and the background, L is length of all borders, and G is mean gradient on the borders). Maximum of MDC defines the segmentation.

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This above equation is used to determine whether a point in the image is an isolated point based on the response magnitude $|R(x, y)|$ and a threshold value T . If the response magnitude is greater than or equal to the threshold, the function returns 1, indicating the presence of an isolated point; otherwise, it returns 0. This helps in the effective detection and segmentation of isolated points in the image.

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Level-set methods The level-set method was initially proposed to track moving interfaces by Dervieux and Thomasset in 1979 and 1981 and was later reinvented by Osher and Sethian in 1988. This has spread across various imaging domains in the late 1990s. It can be used to efficiently address the problem of curve/surface/etc. propagation in an implicit manner. The central idea is to represent the evolving contour using a signed function whose zero corresponds to the actual contour. Then, according to the motion equation of the contour, one can easily derive a similar flow for the implicit surface that when applied to the zero level will reflect the propagation of the contour. The level-set method affords numerous advantages: it is implicit, is parameter-free, provides a direct way to estimate the geometric properties of the evolving structure, allows for change of topology, and is intrinsic. It can be used to define an optimization framework, as proposed by Zhao, Merriman and Osher in 1996. One can conclude that it is a very convenient framework for addressing numerous

applications of computer vision and medical image analysis. Research into various level-set data structures has led to very efficient implementations of this method.